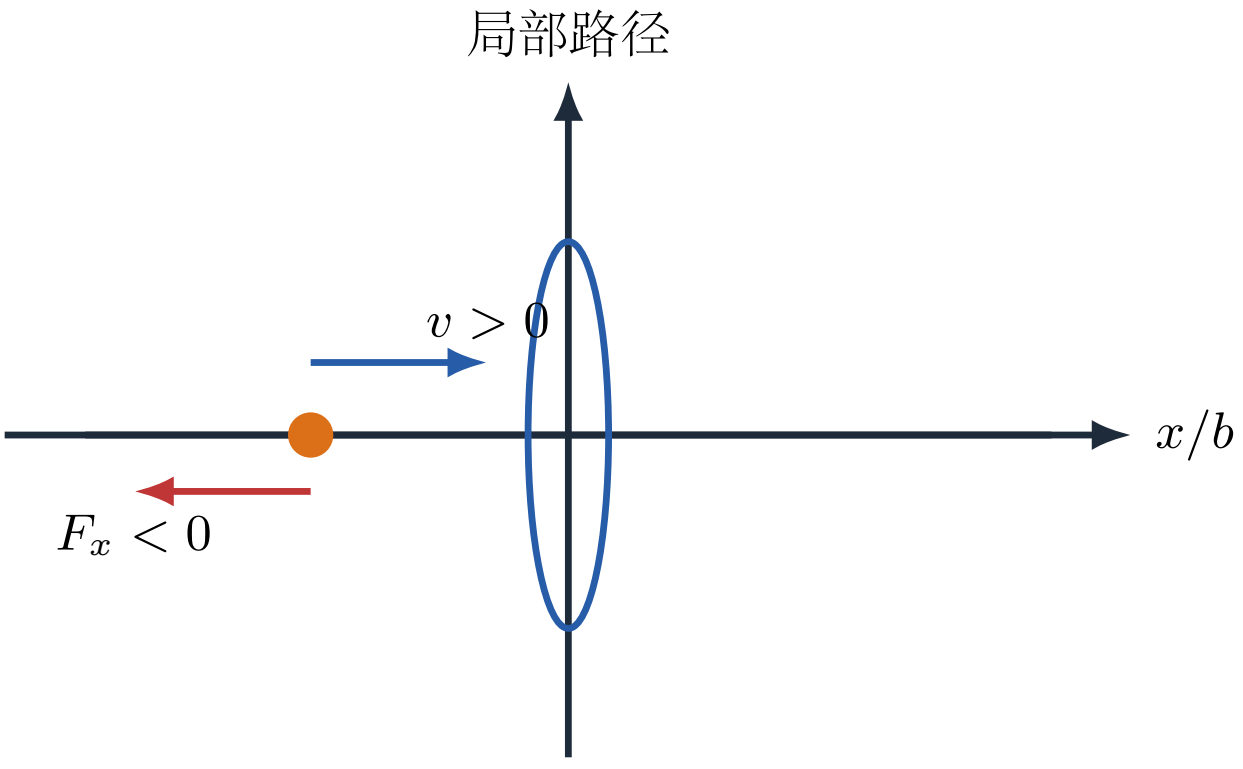

$$F_x = -K^2 v/[R(x^2 + b^2)^3]$$

$$\Delta v = 3\pi\mu_0^2 q^2/(32mRb)$$

完整通过要求 $v_{\text{in}} > \Delta v$

$$K = \mu_0 q b^2/2$$